Product Requirements Document

NPT Intelligence Dashboard for Drilling Operations

Digital solution for analyzing and minimizing Non Productive Time across offshore drilling rigs

Item	Details
Prepared For	ONGC Drilling and Digital Solution Stakeholders
Source Workbook	Copy of ONGC_NPT_Workshop_Dataset.xlsx
Reference Dataset	Synthetic ONGC Drilling NPT Workshop Dataset, Jan-Jun 2024
Technology Direction	Django, PostgreSQL, Plotly, Pandas, Celery, Redis, Docker
Document Purpose	Concise PRD for MVP planning and stakeholder alignment

Executive Summary

The proposed NPT Intelligence Dashboard will convert Excel-based NPT records into an enterprise-grade analytics application. The MVP will focus on visibility, Pareto-based prioritization, drill-down analysis, and management-ready reporting. Later phases may extend the platform into predictive NPT intelligence and workflow-based intervention tracking.

1. Product Vision

To develop a digital NPT Intelligence Dashboard that enables drilling, operations, maintenance, logistics, HSE, and management teams to identify, analyze, prioritize, and reduce Non Productive Time across offshore drilling rigs through data-driven insights.

The solution will transform raw NPT event records into actionable dashboards, trend analytics, root-cause visibility, and intervention recommendations for minimizing rig downtime and improving drilling efficiency.

2. Business Problem

Drilling operations experience recurring NPT due to Weather/Environment, Equipment Failure, Formation/Geological conditions, Well Control events, Logistics/Supply delays, HR/Crew issues, and Regulatory interruptions. The current data is available in Excel but is not yet structured as a dynamic decision-support system.

Reference Dataset Baseline

The workbook Summary_Stats sheet shows 545 NPT events and 8,984.9 total NPT hours during Jan-Jun 2024. Weather/Environment contributes 3,337.4 hours or 37.1 percent of total NPT, followed by Equipment Failure with 2,003.5 hours and Formation/Geological with 1,568.9 hours.

3. Product Objective

- Identify top NPT contributors by cause category, cause detail, rig, contractor, well, phase, and month.
- Detect recurring pain points across drilling phases and rigs.
- Prioritize interventions based on Pareto analysis.
- Enable management review through clear KPIs and visual trends.
- Create a foundation for future predictive analytics and early-warning NPT prevention.

4. Target Users

User Group	Primary Need
Drilling Engineer	Identify top NPT causes and affected drilling phases.
Rig Manager	Monitor rig-wise NPT and recurring issues.
Maintenance Team	Track equipment failure patterns and failure-prone assets.
Logistics Team	Analyze supply-chain delays by material, well, contractor, and phase.
HSE and Well Control Team	Review well-control and safety-critical events.
Senior Management	View fleet-wide KPIs, trends, and operational pain points.

5. Dataset Scope

The current workbook includes NPT event records with date, rig name, well name, contractor, NPT hours, cause category, cause detail, drilling phase, month, and month number. It also includes data dictionary, cause reference, summary statistics, workshop guide, and rig profile sheets for validation and context.

Current Workbook Sheet	Use in PRD and Dashboard
NPT_Data	Primary transactional event dataset for dashboard and filters.
Data_Dictionary	Defines column purpose, accepted values, and examples.
Cause_Reference	Maps cause details to categories, typical duration ranges, and responsible party.
Summary_Stats	Validation baseline for KPIs and chart totals.

Rig_Profiles	Context for rig-wise interpretation and comparison.

6. Key Insights from Current Dataset

Insight Area	Observation
Total NPT	8,984.9 hours across 545 events.
Top Cause Category	Weather/Environment, 3,337.4 hours, 37.1 percent.
Second Major Cause	Equipment Failure, 2,003.5 hours, 22.3 percent.
Third Major Cause	Formation/Geological, 1,568.9 hours, 17.5 percent.
Highest NPT Rig	Sagar Bhushan, 2,276.3 hours.
Peak NPT Month	Jun-2024, 2,616.7 hours.
Major Spike	Jun-2024 shows +122.6 percent month-on-month change.

7. Core Product Features

Feature	Description
Executive KPI Dashboard	Total NPT hours, events, average NPT per event, worst rig, top cause category, and peak month.
Pareto Analysis	Top contributors by cause category, cause detail, rig, contractor, and phase.
Trend Analytics	Monthly NPT trend, rig-wise trend, cause-wise trend, month-on-month movement, and spike detection.
Drill-down Analysis	Interactive filtering by rig, well, contractor, cause category, cause detail, drilling phase, month, and date range.
Rig Performance View	Total NPT, events, average per event, top cause, and rig ranking by NPT.
Cause Intelligence	Total NPT, event count, average NPT, affected rigs and phases, and responsible party mapping.
Pain Point Summary	Auto-generated summary of top causes, rigs needing attention, recurring phases, and abnormal months.

8. Functional Requirements

ID	Requirement
FR-01	Upload Excel or CSV NPT data.
FR-02	Validate required columns before processing.
FR-03	Convert date fields and derive month and month number automatically.
FR-04	Display executive KPIs.
FR-05	Generate Pareto charts for top NPT contributors.
FR-06	Provide filters for rig, contractor, cause, phase, well, month, and date.
FR-07	Display raw filtered data table.
FR-08	Export filtered data and management summaries.
FR-09	Show warning if no data matches selected filters.
FR-10	Provide role-based access for admin, engineer, and viewer.

9. Non-Functional Requirements

Area	Requirement
Performance	Dashboard should load filtered results quickly for operational review.
Usability	Simple layout with clear KPI cards, filters, and interactive charts.
Security	Authenticated login, role-based access, and audit trail.
Reliability	Should not crash for missing values, invalid uploads, or empty filters.
Scalability	Must support expansion from workshop Excel to enterprise database.
Maintainability	Modular Django code with reusable analytics services.
Data Quality	Column validation, missing value checks, and duplicate checks.

10. Suggested Technology Stack

Layer	Recommended Stack
Frontend	Django Templates initially, React later if required.
Backend	Django.
Database	PostgreSQL.
Analytics	Pandas, NumPy.
Visualization	Plotly, Chart.js, or Apache ECharts.
API Layer	Django REST Framework.
Background Jobs	Celery with Redis.
Authentication	Django Auth, LDAP or Azure AD integration if required.
Deployment	Docker, Nginx, Gunicorn.
Reporting	Excel and PDF export using openpyxl and ReportLab.
Future BI	Power BI embedded or direct PostgreSQL connection.
Future ML	scikit-learn, XGBoost, MLflow.

11. Proposed System Architecture

1. User uploads NPT Excel or CSV file.
2. Django backend validates schema and stores clean data in PostgreSQL.
3. Analytics service computes KPIs, trends, Pareto ranking, and pain point summaries.
4. Dashboard frontend displays interactive charts and filters.
5. Users export filtered datasets and management reports.
6. Future ML module predicts likely NPT risk based on historical and contextual factors.

12. Database Design

12.1 Core Table: npt_event

Field	Type	Purpose
id	AutoField / UUID	Primary key.
event_date	Date	Date of NPT occurrence.

rig_name	VARCHAR	Drilling rig name.
well_name	VARCHAR	Well identifier.
contractor	VARCHAR	Responsible contractor or ONGC Own.
npt_hours	DECIMAL	Duration of NPT event.
cause_category	VARCHAR	Top-level NPT category.
cause_detail	VARCHAR	Detailed NPT cause.
drilling_phase	VARCHAR	Phase during event.
month	VARCHAR	Derived display month.
month_num	INTEGER	Derived month sort order.
uploaded_by	User FK	Uploader.
created_at	DateTime	System timestamp.

12.2 Reference Tables

Table	Purpose
rig_master	Rig profile, type, location, asset, and primary contractor.
cause_master	Cause category, cause detail, responsible team, and standard mitigation action.
contractor_master	Contractor profile and service category.
phase_master	Standard drilling phase list.
upload_log	Uploaded file, validation status, user, timestamp, and duplicate flag.
user_master	Role, department, and access level.

13. Analytics and Dashboard Modules

Module	Description
KPI Engine	Computes total NPT, events, average NPT, worst rig, and top cause.
Pareto Engine	Ranks top contributors by category, detail, rig, contractor, and phase.
Trend Engine	Calculates monthly movement, rig trend, and month-on-month change.
Rig Intelligence	Provides rig-wise performance, ranking, and top issue.
Cause Intelligence	Provides root cause patterns and responsible party mapping.
Phase Analysis	Analyzes NPT by drilling phase.
Contractor Analysis	Reviews contractor-wise NPT contribution.
Pain Point Generator	Auto-generates priority action areas for management review.

14. Success Metrics

Metric	Current State	Target
Time to identify top NPT contributor	Manual Excel analysis	< 10 seconds
Monthly NPT review preparation time	Manual compilation	< 15 minutes
Effort in analyzing NPT trends	Manual	70 percent reduction

Visibility of top NPT causes	Limited	100 percent visibility
Identification of recurring NPT patterns	Reactive	Proactive
Data-driven intervention tracking	Not available	Available from Phase 2
Dashboard adoption by drilling teams	Not established	> 80 percent active users
Reduction in controllable NPT	Baseline to be established	10-15 percent reduction in Year 1
Management reporting effort	Manual	One-click export
Data quality validation	Manual	Automated validation every upload

15. User Stories

ID	Persona	User Story
US-01	Executive Management	As a Drilling Asset Manager, I want to view fleet-wide NPT KPIs, top contributing causes, and monthly trends so that I can prioritize intervention areas and track performance across rigs.
US-02	Rig Manager	As a Rig Manager, I want to analyze rig-wise NPT patterns and recurring causes so that I can plan corrective actions and improve operational efficiency.
US-03	Drilling Engineer	As a Drilling Engineer, I want to identify the highest contributing NPT causes by drilling phase so that I can focus preventive measures during upcoming operations.
US-04	Maintenance Engineer	As a Maintenance Engineer, I want to monitor equipment-related failures across rigs and contractors so that I can improve maintenance planning and reduce breakdowns.
US-05	Logistics Team	As a Supply Chain Coordinator, I want to analyze logistics-related delays and affected rigs so that I can improve material availability and contractor coordination.
US-06	HSE and Well Control Team	As a Well Control Engineer, I want to review well control incidents and associated NPT hours so that I can identify recurring risks and strengthen preventive controls.
US-07	Management Review Committee	As a Reviewer, I want the system to automatically highlight top pain points and recommended interventions so that I can take faster decisions during review meetings.

16. Acceptance Criteria

ID	Acceptance Criteria
AC-01	Dashboard displays Total NPT Hours, Total NPT Events, Average NPT/Event, Worst Performing Rig, and Top Cause Category.
AC-02	For current reference data, system correctly displays 8,984.9 total NPT hours.
AC-03	For current reference data, system identifies Weather/Environment as the highest NPT contributor.
AC-04	Users can filter by rig, well, contractor, cause category, cause detail, phase, month, and date range.
AC-05	All KPIs and charts refresh dynamically after filters are applied.
AC-06	Dashboard does not crash when filter results are empty.
AC-07	System displays: No Data Available for Selected Filters when no records match.
AC-08	System provides Pareto analysis for cause category, cause detail, contractor, rig, and phase.
AC-09	System generates monthly trends and rig-wise comparisons.
AC-10	Users can identify top 10 NPT contributors within 10 seconds.
AC-11	System validates uploaded Excel files against mandatory schema.

AC-12	Invalid files return meaningful validation messages.
AC-13	Duplicate uploads are detected and flagged.
AC-14	Users can export filtered data to Excel.
AC-15	Users can export dashboard summaries to PDF.
AC-16	System generates printable Management Review reports.
AC-17	Only authenticated users can access the dashboard.
AC-18	Role-based access is supported.
AC-19	Upload history is maintained.

17. Data Schema

17.1 Master Dataset: NPT Events

Field Name	Type	Mandatory	Description
id	UUID / Auto	Yes	Primary key.
event_date	Date	Yes	Date of NPT occurrence.
rig_name	VARCHAR(100)	Yes	Rig name.
well_name	VARCHAR(100)	Yes	Well identifier.
contractor	VARCHAR(100)	Yes	Contractor or ONGC Own.
npt_hours	Decimal	Yes	NPT duration in hours.
cause_category	VARCHAR(100)	Yes	Top-level NPT category.
cause_detail	VARCHAR(200)	Yes	Specific root cause.
drilling_phase	VARCHAR(100)	Yes	Operational phase.
month	VARCHAR(20)	Yes	Derived month label.
month_num	Integer	Yes	Sorting index for trend charts.
uploaded_by	User FK	Yes	Upload owner.
uploaded_on	Timestamp	Yes	Upload timestamp.
remarks	Text	No	Additional comments.

17.2 Cause Categories in Current Dataset

- Equipment Failure
- Formation/Geological
- Weather/Environment
- HR/Crew
- Logistics/Supply
- Well Control
- Third Party/Regulatory

18. Proposed Master Tables

Master Table	Recommended Columns
Rig Master	Rig ID, Rig Name, Rig Type, Asset, Location, Contractor.

Cause Master	Cause Category, Cause Detail, Responsible Team, Standard Mitigation Action.
Contractor Master	Contractor Name, Contract Type, Service Category.
User Master	User, Role, Department, Access Level.
Upload Log	File Name, Upload Date, Uploaded By, Status, Validation Remarks.

19. Initial Pain Points and Digital Interventions

Pain Point	Suggested Digital Intervention
Weather/Environment is the largest NPT contributor.	Add weather trend correlation and weather early-warning dashboard in later phase.
Equipment Failure is a major controllable contributor.	Build equipment failure Pareto and maintenance action tracker.
Sagar Bhushan has highest NPT.	Create rig-specific deep-dive dashboard.
Jun-2024 shows a major NPT spike.	Add monthly anomaly detection.
Logistics delays are recurring.	Track supply delays by item, contractor, well, and phase.
HR/Crew causes are frequent but lower duration.	Monitor crew change, training, sickness, competency, and shift handover delays.
Excel-based review is static.	Move to PostgreSQL-backed dynamic dashboard.

20. Recommended MVP Screen Layout

7. Header: ONGC NPT Intelligence Dashboard.
8. Filter panel: Date, Rig, Contractor, Cause Category, Cause Detail, Phase, Well, Month.
9. KPI row: Total NPT Hours, Events, Avg NPT, Worst Rig, Top Cause, Peak Month.
10. Charts row 1: NPT by Cause Category and NPT by Rig.
11. Charts row 2: Monthly Trend and Top Cause Details Pareto.
12. Drilldown section: Rig-wise, contractor-wise, and phase-wise table.
13. Pain point summary: Auto-generated insights and priority actions.
14. Export section: Download Excel and PDF.

21. Implementation Roadmap

Phase	Scope
Phase 1: MVP Dashboard	Excel upload, validation, KPI cards, Pareto charts, monthly trends, rig/cause filters, raw data table, Excel export.
Phase 2: Operational Analytics	PostgreSQL historical data, rig benchmarking, contractor analytics, phase heatmap, pain point summary, PDF report.
Phase 3: Management Dashboard	Role-based dashboards, executive view, monthly review pack, automated email report, upload audit trail.
Phase 4: Predictive NPT Intelligence	NPT risk scoring, pattern detection, early warning indicators, recommendation engine, integration with DDR, maintenance, logistics, and weather data.

22. Out of Scope for Phase 1 MVP

Area	Excluded from MVP
Technology	Mobile application, offline mode, real-time sensor integration, IoT device integration, SAP integration, ERP integration.

Advanced Analytics	AI-based root cause prediction, predictive NPT forecasting, digital twin modeling, equipment remaining life prediction, production loss estimation.
External Integrations	Weather API integration, contractor ERP integration, DGMS systems integration, CMMS integration.
Workflow Automation	Automatic work order generation, maintenance scheduling, inventory procurement automation, contractor performance management.

23. Future Release Candidates

- Predictive NPT analytics and risk scoring.
- AI recommendation engine for preventive action.
- Early warning system for high-risk rigs and phases.
- Action tracking and accountability module.
- Lessons learned repository linked to cause categories.
- Integration with Daily Drilling Reports, maintenance, logistics, and weather data.
- Power BI executive cockpit.
- GenAI-based NPT root cause summarization.

24. Final Recommendation

Start with a Django and PostgreSQL based MVP rather than a standalone dashboard because the long-term requirement is not only visualization but also structured data storage, access control, upload governance, historical benchmarking, audit trail, reporting, and future ML-based NPT prediction.

For quick demonstration, a Streamlit prototype can be used. For an enterprise-grade ONGC digital solution, Django, PostgreSQL, Plotly, Pandas, Celery, Redis, Docker, and optional Power BI integration should be the preferred stack.

Recommended PRD Success Statement

The NPT Intelligence Dashboard shall enable ONGC drilling teams to identify the top contributors to Non Productive Time within 10 seconds, reduce manual analysis effort by at least 70 percent, provide clear visibility into recurring operational bottlenecks, and create the foundation for AI-driven NPT prediction and prevention.

25. Source Note

This PRD is based on the attached workbook: Copy of ONGC_NPT_Workshop_Dataset.xlsx. Quantitative baselines were taken from the Summary_Stats sheet and schema references were taken from the Data_Dictionary and Cause_Reference sheets. The workbook states that the data is synthetic and generated for workshop purposes, and should not be used for operational decisions.